

Jensen Peng

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‹ Software Engineer /›

Contact

Phone
+61 413993029
+86 15776692817

Email
pjtpengjingtong@gmail.com
pengjingtong@pku.edu.cn

Address
Perth, WA · Open to relocation

Github
https://github.com/ArcueidShiki

Linkedin
https://www.linkedin.com/in/jingtong-peng-3068672b6/

Leetcode
https://leetcode.cn/u/arcueidshiki/

skills

- C/C++
- Bash
- JavaScript
- Python
- Java
- TypeScript
- Linux
- Windows
- Redis
- SpringBoot
- Flask
- Threejs
- Cloudflare
- Swift
- React
- Nextjs
- Vue
- HTML
- CSS
- QT
- MySQL
- Sqlite
- MongoDB
- Git
- Docker
- AWS
- Arduino
- Raspberry Pi

Education

The University of Western Australia
Master of Information Technology
2024-02-14 - 2025-12-17

Peking University
Master of Pharmacy
2019-09-01 - 2021-09-09

Harbin Medical University
Bachelor of Pharmacy
2015-09-01 - 2019-07-03

Language

- English (fluent)
- Chinese (native)

Work Experience

Indigo Australasia — Full Stack Developer

Nov 2025 – Present

- Deliver end-to-end .NET, Angular and PostgreSQL features for Get2Go and SAH, supporting assistive technology distribution.
- Optimise product queries with CPU profiling, flamegraphs and SQL tuning; maintain operational data.
- Integrate OpenSearch indexes to accelerate fuzzy product search and autocomplete.
- Automate business workflows and data synchronisation with Power Automate.
- AWS hackathon: resolve AWS–Azure Data Factory integration using prompt engineering, CSV-to-JSON Schema conversion and pipeline implementation.
- Automate batch recognition and information extraction from supplier invoices.
- Internal hackathon: build a Claude-assisted financial accounting prototype for finance data, payroll workflows and email automation.

Huawei Technologies Co., Ltd. — System Engineer

2022-09-06 - 2024-02-14

Hiperf（Performance analysis tool)

- Developed core modules of Hiperf, a cross-platform performance profiler for HarmonyOS with Android and Linux compatibility.
- Performance Data Collection: Implemented CPU-cycle and instruction sampling with perf_event_open, mmap and ring buffers, supporting low-level performance analysis.
- Call-Stack Analysis: Built C++/JavaScript stack unwinding using DWARF debug information and stack pointers to identify performance bottlenecks.
- Symbol Resolution: Parsed shared libraries and compressed debug information to map sampled addresses to functions and source-level context.
- Concurrent Data Exchange: Implemented shared-memory buffers and asynchronous processing, with cross-platform library interfaces for report generation.
- Command-Line Tooling: Developed help, list, dump and record commands covering collection, statistics and reporting workflows.
- Runtime Context: Maintained process and thread snapshots to preserve sample traceability and call-stack associations.
- Toolchain Troubleshooting: Collaborated with an international cross-compilation team to investigate LLVM compiler and kernel issues.
- Performance Optimisation: Improved cache-page mapping and report read/write paths to reduce CPU overhead and accelerate report generation.
- Engineering Quality: Wrote automated CPU/memory tests, build scripts and design documentation; supported technical reviews and evaluated Simpleperf and Perfetto.

Project Experience

US Stock Paper Trading Platform — Full Stack Dev

2025-04-03 - 2025-05-10

- Designed and built a US stock paper-trading platform combining market visualisation, portfolio management, financial analysis and user chat.
- Architecture & APIs: Developed Flask REST APIs and WebSocket communication to connect trading workflows, financial data and live interface updates.
- Frontend Delivery: Built responsive Bootstrap interfaces and ECharts dashboards, with jQuery for dynamic interactions.
- Data Modelling: Designed a SQLite schema for users, portfolio/financial records and chat history.
- Performance: Refined data queries and WebSocket message flows to support responsive updates and a consistent user experience.

Options & Indicators Lab — Full Stack Dev

Independent project

Designed and built a bilingual US-market options and indicators strategy lab, translating financial models into interactive decision-support tools. Features include multi-leg strategy comparison, payoff visualisation and technical-indicator experiments.

- Technical Highlights — Options & Indicators Strategy Lab:
- Financial Modelling: Implemented Black–Scholes valuation and Greek decomposition, connecting model assumptions to pricing and risk visualisations.
- Interactive Analysis: Built expiry and T+0 payoff curves, breakeven analysis and multi-leg strategy comparisons to explore scenario trade-offs.
- Strategy Screening: Added filters for market direction, premium paid/received and risk structure, making a broad strategy library easier to navigate.
- Indicator Experiments: Developed a companion chart lab and indicator-combination matrix using synthetic OHLCV data for reproducible educational experiments.
- Product Delivery: Integrated bilingual explanations and interactive Canvas visualisations into a browser-based learning experience.

Remote Control Software

Full Stack Dev

2024-12-01 - 2025-01-15

- Designed a Windows remote-assistance application for desktop sharing and control.
- Protocol Design: Defined packet headers, lengths, commands and checksums for a custom TCP communication protocol.
- Client Delivery: Built MFC modules for the UI, networking, streaming, desktop capture and remote input handling.
- Desktop Streaming: Captured desktop state and transmitted screen updates over TCP socket streams.
- Remote Input: Implemented mouse and keyboard event handling to support interactive assistance.
- Maintainability: Refactored the client into an MVC structure, separating presentation, business logic and data models.

Penni Part-time Job Platform

cfc

Full Stack Dev

2024-06-18 - 2024-07-16

- Contributed frontend features to a Coders for Causes platform helping retirees find part-time work.
- User Flows: Implemented registration/login, role switching, validation and error handling.
- Responsive UI: Built Flexbox/Grid layouts and a progress indicator for multi-step registration.
- Reusable Forms: Centralised field state and validation in reusable form components.
- Media Uploads: Integrated webcam capture and file uploads with client-side preview and cropping.
- Authentication: Used Axios interceptors to attach JWTs and manage authenticated API calls.
- Data Fetching: Applied React Query caching to reduce redundant requests and keep UI data consistent.

Internship

St John of God Health Care

ST JOHN OF GOD Health Care

Web / UX Intern

2025-06-18 - 2025-07-16

- Audited internal and public websites with Google Lighthouse, assessing performance, accessibility, SEO and implementation quality.
- Analysed Google Analytics data and mapped user journeys to identify navigation friction and conversion barriers.
- Translated findings into actionable usability and WCAG accessibility recommendations for digital and communications teams.
- Created Figma prototypes to validate proposed website changes against user needs and brand requirements.
- Nominated by my mentor for the Outstanding Intern Award.

Hackathons

- Attended the Quantum Bit AI Developer Conference, 2019
- Participated in the Huawei HDC Conference, 2023
- UWAYE Presents: **Rio Tinto** X **BCG** Hackathon 2024
- Coders for Causes: winter project 2024
- **Susquehanna** Algothon 2025
- WeMoney Financial Wellness AI Hackathon 2025
- CFC **Anthropic** X **Jane Street** Hackathon 2025

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Professional Profile

Full-stack software engineer combining C++ systems and performance tooling experience at Huawei with end-to-end C# .NET, Angular and PostgreSQL delivery at Indigo Australasia. Experienced in API and database design, search optimisation, cloud integration and operational automation, applying AI-assisted development to practical business workflows. Holds a Master of Information Technology from the University of Western Australia.
Based in Perth; open to relocation to Melbourne, Sydney or Brisbane. Seeking employer visa sponsorship.